

$^{34}\text{S}(\text{d},\text{p})$ 2024Ku24,1971Va18,1969Mo12

$J^\pi=0^+$ for ^{34}S g.s.

2024Ku24: An 8-MeV/nucleon deuteron beam was produced from the 9-MV Super-FN tandem Van de Graaff accelerator at FSU.

The target was carbon-backed sulfur enriched in ^{34}S . Protons were spatially dispersed by the Super-Enge Split-Pole Spectrograph and detected using a position-sensitive ionization chamber (ΔE) and a plastic scintillator (E) at the focal plane with FWHM $\approx$ 45 keV. An additional $^{34}\text{S}(\text{d},\text{p}\gamma)^{35}\text{S}$ measurement using the CeBr₃ Array (CeBrA) demonstrator provides information on some overlapping peaks produced from the ^{32}S contaminants in the target. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 23 levels, single-neutron spectroscopic factors and strengths from finite-range PTOLEMY-DWBA analysis.

1971Va18: A 10-MeV deuteron beam was produced from the Utrecht 6-MV tandem accelerator. Targets were 100 $\mu\text{g}/\text{cm}^2$ PbS on carbon plus formvar foils and 5 $\mu\text{g}/\text{cm}^2$ pure ^{34}S embedded in aluminum foil. Protons were detected using one 15-mm long and seven 30-mm long, 0.6-mm thick position-sensitive detectors (PSD) in the focal plane of the Utrecht split-pole magnetic spectrograph. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 17 levels, L-transfers, J, π , spectroscopic factors from DWUCK-DWBA analysis for the ground state and seven excited states of ^{35}S .

1969Mo12: A 6.495-MeV deuteron beam was produced from the ONR-CIT tandem accelerator. The target was an enriched target of 450- $\mu\text{g}/\text{cm}^2$ 25 CdS (85% ^{34}S) evaporated onto a 301- $\mu\text{g}/\text{cm}^2$ 15 gold foil and a natural target of 289- $\mu\text{g}/\text{cm}^2$ 30 Sb²S³ evaporated onto a 289- $\mu\text{g}/\text{cm}^2$ 30 gold foil. Protons were detected using an array of 16 Au-Si surface barrier detectors with FWHM=30 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 37 levels.

1984Pi03: A 12.3-MeV deuteron beam was produced from a cyclotron. The target was CdS (61.9% in ^{36}S , 37.7% in ^{34}S), 100 $\mu\text{g}/\text{cm}^2$ on a 20 $\mu\text{g}/\text{cm}^2$ carbon backing. Protons were analyzed by a multi-angle magnetic spectrograph and detected using 700-mm long nuclear emulsion plates Ilford L4. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 13 levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis for six excited states of ^{35}S .

1979So01: A 3.55-MeV deuteron beam was produced from a Van de Graaff electrostatic generator. The target was 18- $\mu\text{g}/\text{cm}^2$ GeS (80% ^{36}S , 20% ^{34}S) on a 25- $\mu\text{g}/\text{cm}^2$ carbon backing. Protons were analyzed by a magnetic spectrograph and recorded by a spark chamber with FWHM=10-15 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels.

1971Ko33: A 6.6-MeV deuteron beam was produced from a cyclotron. Targets were GeS films of 70-80 $\mu\text{g}/\text{cm}^2$ (98% enrichment in ^{34}S) evaporated onto a 20 $\mu\text{g}/\text{cm}^2$ carbon substrate. Protons were detected using an 800- μm -thick Si(Li). Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from distorted-wave analysis for the ground state and five excited states of ^{35}S .

1971Me12: An 18-MeV deuteron beam was produced from the Yale MP tandem Van de Graaff accelerator. The target was H²S gas of natural isotopic composition (95.06% ^{32}S , 4.18% ^{34}S). Protons were detected using a ΔE -E telescope of silicon surface-barrier detectors. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from local, zero-range JULIE-DWBA analysis of $\sigma(E_{\text{p}},\theta)$ for the ground state and two excited states of ^{35}S .

1966Sc09: 9- and 12-MeV deuteron beams were produced from the Argonne tandem accelerator. The target was PbS. Protons were detected using silicon surface barrier detectors. Measured $\sigma(E_{\text{p}},\theta)$ for the ground state and one excited state of ^{35}S .

1958En51: 6.006- and 6.542-MeV deuteron beams were produced from the MIT-ONR electrostatic generator. Targets were Sb²S³ of natural isotopic constitution (4.2% ^{34}S). Energies of charged reaction products emitted from the target at angles of 50, 90, and 130° were measured by a broad-range magnetic spectrograph and nuclear emulsions. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 5 levels.

Other measurements: **1976We29:** $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ angular distributions for the ground state and the first excited state of ^{35}S .

1973Co25: $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ excitation functions for the two lowest excited states of ^{35}S .

Theoretical calculations: **1977Os07**, **1974Os02**.

 ^{35}S Levels

C²S from **2024Ku24**. The normalization was determined from the L=3 strengths, $G_{L=3}$, summed up through $E_{\text{x}}=5.1$ MeV and divided by the expected 1f_{7/2} vacancy, i.e., $\Sigma G_{L=3}/(2j+1)$, where $j=7/2$. This is equivalent to $\Sigma C^2S_{L=3}=1$ up to 5.1 MeV. For C²S given under comments, $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N/(2J_{\text{f}}+1)\times(2J_{\text{i}}+1)$, where the normalization factor $N=1.53$. The C²S from **1971Me12** based on $N=1.58$ has been renormalized by evaluators using $N=1.53$.

E(level)	J^π &	L ^a	C ² S ^a	Comments
0	3/2 ⁺	2	0.54 12	L: 2 also from 1971Va18 , 1971Ko33 , and 1971Me12 . C ² S: others: 0.43 11 (1971Va18), 0.33 (1971Ko33), and 0.48 (1971Me12).
1571.92 [#] 19	1/2 ⁺	0	0.14 7	L: 0 also from 1971Va18 , 1971Ko33 , and 1984Pi03 .

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d,p})$ [2024Ku24,1971Va18,1969Mo12](#) (continued) ^{35}S Levels (continued)

E(level)	J^π	L^a	C^2S^a	Comments
1991.08 [#] 16	7/2 ⁻	3	0.87 9	C ² S: others: 0.17 4 (1971Va18), 0.25 (1971Ko33), and 0.154 (1984Pi03). C ² S: others: 0.68 17 (1971Va18), 0.38 8 (1971Ko33), 0.65 10 (1971Me12), and 0.841 12 (1984Pi03).
2347.59 [#] 15	3/2 ⁻	1	0.62 7	E(level): others: 2347 8 (1969Mo12) and 2336 10 (1971Va18). L: 1 also from 1971Va18 and 1984Pi03 . C ² S: others: 0.52 13 (1971Va18), 0.52 (1971Me12), 0.33 (1971Ko33), and 0.505 (1984Pi03).
2724 8	5/2 ⁺	2	0.02 1	E(level): weighted average of 2722 8 (1969Mo12) and 2726 8 (1971Va18). L: other: (2,3) from 1971Va18 .
2941 10	3/2 ⁺	2	<0.02	E(level): weighted average of 2943 10 (1969Mo12) and 2939 10 (1971Va18).
3420 8	5/2 ⁺	2	0.02 1	E(level): weighted average of 3422 8 (1969Mo12) and 3415 12 (1971Va18).
3560 8				E(level): weighted average of 3563 8 (1969Mo12) and 3555 9 (1971Va18).
3596 [†] 8				E(level): other: 3595 9 (1971Va18).
3675 [‡] 10	(1/2 ⁻ , 3/2 ⁻)	(1)		J^π : assigned from L=(1) and also adopted in the Adopted Levels. L: from 1971Va18 . C ² S: <2.5×10 ⁻³ for $J^\pi=1/2^-$ and <1.25×10 ⁻³ for $J^\pi=3/2^-$ (1971Va18).
3801.90 [#] 30	3/2 ⁻	1	0.07 4	E(level): others: 3804 8 (1969Mo12) and 3795 10 (1971Va18). L: 1 also from 1971Va18 . C ² S: 0.09 3 (1971Va18).
3885 [†] 10				E(level): other: tentative 3866 10 (1971Va18) and 3907 10 (1971Va18).
4025 10				E(level): weighted average of 4025 10 (1969Mo12) and 4025 10 (1971Va18).
4109 10				E(level): weighted average of 4112 10 (1969Mo12) and 4105 10 (1971Va18).
4189.87 [#] 26	1/2 ⁻	1	0.17 2	E(level): others: 4190 8 (1969Mo12) and 4196 12 (1971Va18). C ² S: other: 0.12 3 (1971Va18).
4305 8				E(level): weighted average of 4302 8 (1969Mo12) and 4312 12 (1971Va18).
4481 [†] 8	3/2 ⁺ , 7/2 ⁻	2+3	0.07, 0.05	E(level): other: 4477 doublet (2024Ku24). J^π : for two components, as given in 2024Ku24 . L: 2,3 doublet (2024Ku24). C ² S: others: 0.07 2 for 3/2 ⁺ component and 0.05 2 for 7/2 ⁻ component (2024Ku24).
4575 [†] 8				
4837 [†] 8				
4903.28 [#] 16	1/2 ⁻	1	0.78 8	E(level): other: 4903 8 (1969Mo12). L: 1 also from 1984Pi03 . C ² S: other: 0.776 (1984Pi03).
4963.12 [#] 16	3/2 ⁻	1	0.32 3	E(level): other: 4965 8 (1969Mo12). L: 1 also from 1984Pi03 . C ² S: other: 0.218 (1984Pi03).
5058 [†] 8	7/2 ⁻	3	0.08 1	
5126 [†] 11				
5342 [†] 8				
5475 [†] 10				Only observed in 1969Mo12 and not included in the Adopted Levels.
5542 [†] 8				
5740 [@] 20	5/2 ⁻ , 7/2 ⁻	3	0.03 1	J^π : assigned from L=3 and also adopted in the Adopted Levels. 2024Ku24 assigns (5/2 ⁻) and states that the J^π assignment is purely speculative.
5890 [@] 20	3/2 ⁺ , 5/2 ⁺	2	0.11 2	J^π : assigned from L=2 and also adopted in the Adopted Levels. 2024Ku24 assigns (3/2 ⁺) and states that the J^π assignment is purely speculative.
5980 [†] 10				
6078.6 [#] 13	1/2 ⁻ , 3/2 ⁻	1		E(level): other: 6080 8 (1969Mo12).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d,p})$ **2024Ku24,1971Va18,1969Mo12 (continued)** ^{35}S Levels (continued)

E(level)	J^π &	L ^a	Comments
J ^π : assigned from L=1 and also adopted in the Adopted Levels. L: from 1984Pi03. C ² S=0.086 for J ^π =1/2 ⁻ and 0.042 for J ^π =3/2 ⁻ .			
6292 [†] 8			
6334 [†] 8			
6344 [†] 8			
6446 [†] 8			
6496 [†] 8			
6537.7 [#] 14			
6545.1 [#] 13			E(level): other: 6543 8 (1969Mo12).
6584 [†] 10			
6635.2 [#] 13		(2,3)	E(level): others: 6634 8 (1969Mo12) and 6625 30 (2024Ku24).
6677 [†] 8			
6891.3 [#] 14	(1/2 to 7/2) ⁽⁻⁾	(1,3)	E(level): others: 6892 10 (1969Mo12) and 6915 30 (2024Ku24).
7021 10		(2,3)	E(level): weighted average of 7022 10 (1969Mo12) and 7005 35 (2024Ku24).
7150 [@] 35		(0,3)	
7205 [@] 35		(1,2)	
7482.7 [#] 13		(1,2)	E(level): other: 7495 40 (2024Ku24).
7570 [@] 40		(2,3)	
7640 [@] 40		(2,3)	

[†] From 1969Mo12.[‡] From 1971Va18.[#] From 1984Pi03.[@] From 2024Ku24.& Assumed for extracting C²S, unless otherwise noted.^a From PTOLEMY-DWBA analysis of the measured $\sigma(\theta)$ in 2024Ku24, unless otherwise noted. $\gamma(^{35}\text{S})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
1572	1571.92	1/2 ⁺	0	3/2 ⁺	E_γ : from 2024Ku24.
1991	1991.08	7/2 ⁻	0	3/2 ⁺	E_γ : from 2024Ku24.

 $^{34}\text{S}(\text{d,p})$ 2024Ku24,1971Va18,1969Mo12

Level Scheme